1 Cycle-slip detection and repair

The uncombined filter (the observation-equations section of the Foundations document) never forms a linear combination on the measurement-update path: per-signal $P^i_{r,s}$ and $\Phi^i_{r,s}$ enter the uncombined code and phase observation equations directly. Cycle slips are instead screened by a bank of *diagnostic* combinations, each constructed so that the geometric range ρ^i_r, the clocks $c(\mathrm{d}t_r - \mathrm{d}t^i)$ and the troposphere T^i_r cancel, leaving an observable whose only between-epoch variation is measurement noise plus any integer carrier discontinuity. The active detectors are Melbourne–Wübbena (`src/core/slip/MwSlipDetector.cpp`, the primary) and GRAPHIC (`src/core/slip/GraphicSlipDetector.cpp`); the extra-wide-lane detector (`src/core/slip/EwlSlipDetector.cpp`) runs when a third frequency is present. The geometry-free and ionosphere-free phase detectors (`src/core/slip/GfSlipDetector.cpp`, `src/core/slip/IfPhaseSlipDetector.cpp`) are staged — present in the tree but not compiled into the current build — and their algebra is recorded here for completeness. The bank is composed at run time from the config block `estimation.cycle_slip` (Section 1.1).

Throughout this section we suppress the receiver index r (single-receiver PPP / PPP-RTK) and write $f_1 > f_2$ for the two carriers of a band pair, sorted by frequency so that $f_1 - f_2 > 0$ and $\lambda_1 < \lambda_2$ (`src/core/slip/MwSlipDetector.cpp`, the band-frequency sort). Carrier phase is read from RINEX in cycles and scaled by $\lambda_s = c/f_s$ before combining, so Φ^i_s below is always in metres, consistent with the phase observation equation. We reserve the operator $\delta\cdot$ for a *temporal* integer slip step (e.g. δN_s is the change in the ambiguity on signal s across the slip epoch); this is deliberately distinct from the between-satellite single difference $\Delta\cdot$ of the differencing section of the Foundations document, which also uses Δ.

1.1 Detector registry and composition

The detectors are not hard-wired. Each implements a narrow polymorphic contract `ISlipDetector` (`include/slip/ISlipDetector.hpp`) — a single per-epoch entry point

$$\texttt{detect}(\text{SlipEpochContext}) \;\longrightarrow\; \{\,\text{SlipEvent}\,\},$$

plus lifecycle hooks (`resetSatellite`, `resetSatelliteSignal`, `enableTracing`, `setCorrections`). A `SlipEvent` is the union of the legacy per-detector result structs; its `granularity` field distinguishes a *per-signal* flag (GRAPHIC, GF, IF: one carrier) from a *band-difference* flag (MW, EWL: a wide-lane integer $\delta N_{\mathrm{WL}} = \delta N_1 - \delta N_2$ that, on its own, cannot attribute the jump to one signal — the two-unknown problem Equation (12) resolves downstream). Inputs are bundled in `SlipEpochContext` so the signature is uniform even though only the staged IF detector consumes the extra 3D position sigma and receiver ECEF it carries.

`SlipDetectorRegistry::createEnabled` (`src/core/slip/SlipDetectorRegistry.cpp`) builds the config-selected set from `estimation.cycle_slip.{mw,graphic,ewl,gf,if}` in a deterministic order — MW first (it is the primary and the sole source of the wide-lane integers consumed by ambiguity resolution (the ambiguity-resolution document) and the BSSD pivot machinery (`src/core/filter/PivotManager.cpp`)), then the auxiliaries. MW defaults on; the auxiliary detectors default off, preserving the historical MW-only behaviour. A kind that is enabled in config but *unavailable* in this build — the staged `gf` and `if` detectors, whose `.cpp` is not compiled — is skipped with a warning rather than aborting, so a forward-looking config degrades gracefully. `CycleSlipManager` drives the whole set through one loop and recovers the

MW detector's richer API (WL-lock queries, pivot-handover hints, per-satellite quality scoring) through a typed down-cast (`asMelbourneWubbena`) rather than widening the base interface.

`turbo_edit` (Section 1.3) sits a level above the registry: it is *not* a detector kind but a *coordination mode* that, when the MW detector is enabled, pairs the MW wide-lane integer with the geometry-free phase jump to recover per-signal integers. It is therefore documented separately from the detector bank below.

1.2 Test-statistic combinations

1.2.1 Melbourne–Wübbena (wide-lane)

The Melbourne–Wübbena (MW) observable is the wide-lane carrier phase minus the narrow-lane code. Introducing the wide- and narrow-lane wavelengths

$$\lambda_{\mathrm{WL}} \;:=\; \frac{c}{f_1 - f_2}, \qquad \lambda_{\mathrm{NL}} \;:=\; \frac{c}{f_1 + f_2}, \tag{1}$$

the engine forms (`MwSlipDetector::computeMwMeters`)

$$\boxed{\mathrm{MW}^i \;=\; \underbrace{\frac{f_1\,\Phi_1^i - f_2\,\Phi_2^i}{f_1 - f_2}}_{\text{wide-lane phase}} \;-\; \underbrace{\frac{f_1\,P_1^i + f_2\,P_2^i}{f_1 + f_2}}_{\text{narrow-lane code}}}. \tag{2}$$

The range-additive terms common to the uncombined code and phase observation equations — geometry, clocks, troposphere — cancel between the two bracketed combinations. The slant ionosphere also cancels: the wide-lane phase carries the ionospheric coefficient $-(f_1\mu_1 - f_2\mu_2)/(f_1 - f_2)$ and the narrow-lane code carries $+(f_1\mu_1 + f_2\mu_2)/(f_1 + f_2)$, and with $\mu_s = f_1^2/f_s^2$ (the ionosphere-scaling relation) these two coefficients are equal, so the difference (2) removes the ionosphere exactly to first order. What remains is a near-constant equal to the wide-lane ambiguity plus hardware bias,

$$\mathrm{MW}^i \;\approx\; \lambda_{\mathrm{WL}}\, N_{\mathrm{WL}}^i \;+\; b_{\mathrm{MW}}^i \;+\; \varepsilon_{\mathrm{MW}}, \qquad N_{\mathrm{WL}}^i = N_1^i - N_2^i, \tag{3}$$

so a wide-lane slip $\delta N_{\mathrm{WL}} = \delta N_1 - \delta N_2$ appears as a step of $\lambda_{\mathrm{WL}}\,\delta N_{\mathrm{WL}}$. For GPS L1/L2, $\lambda_{\mathrm{WL}} \approx 0.861\,\mathrm{m}$ while the dominant error is the metre-level narrow-lane *code* noise; with code noise $\sigma_P \sim 0.3\,\mathrm{m}$ the MW noise is $\sigma_{\mathrm{MW}} \approx \sigma_P/(\sqrt{2}\,\lambda_{\mathrm{WL}}) \sim 0.25$ wide-lane cycles, i.e. code-noise-limited. The integer-ready (BIA-corrected) variant subtracts the satellite MW bias b_{MW}^i, which has the same algebra as Equation (2) applied to the OSB phase and code biases of the uncombined code and phase observation equations.

The level series $\mathrm{MW}^i(k)$ is screened against a robust adaptive gate whose deviation statistic is configurable via `estimation.cycle_slip.mw_detect_reference`. The recommended — and replay-default — choice is the first time difference $\mathrm{TD1}(k) = \mathrm{MW}^i(k) - \mathrm{MW}^i(k-1)$, gated against the median-absolute-deviation of its own history; the second time difference $\mathrm{TD2}(k) = \mathrm{MW}^i(k) - 2\,\mathrm{MW}^i(k-1) + \mathrm{MW}^i(k-2)$ is available (`mw_td_order`) for curvature immunity at a $\sqrt{6}$ noise penalty versus the $\sqrt{2}$ of TD1 (the white-noise scaling of each statistic is the single source of truth in `MwSlipDetector::activeStatScale`). A windowed local median and the legacy Hatch-mean reference are retained for A/B comparison (`local_median`, `hatch_mean`). The threshold itself is fully data-driven — no stochastic-model σ enters — combining a per-arc MAD, an EWMA-smoothed cross-satellite MAD floor and an elevation $1/\sin\theta$ shape, with a breadth cascade (band-pair → constellation → global-no-GLONASS) for sparse or cold-start arcs; the empirical construction lives in `MwSlipDetector::computeAdaptiveThreshold` and is intentionally not reproduced here.

1.2.2 Extra-wide-lane (EWL)

For three-or-more frequency satellites the same MW algebra (Equation (2)) is applied to a pair of *non-L1* carriers (`EwlSlipDetector::computeEwlMwMeters`; band digit ≥ 2). Because the two carriers are spectrally close — $f_1 - f_2$ small — the extra-wide-lane wavelength

$$\lambda_{\mathrm{EWL}} \;=\; \frac{c}{f_1 - f_2} \tag{4}$$

is very long: GPS L2/L5 gives $\lambda_{\mathrm{EWL}} \approx 5.86\,\mathrm{m}$ and Galileo E5a/E5b $\approx 9.77\,\mathrm{m}$. Metre-level code noise then maps to $\lesssim 0.05$ EWL cycles, so the EWL integer $N_{\mathrm{EWL}} = N_2 - N_5$ resolves in a single epoch and a slip δN_{EWL} is sized with near-certainty. Together with the L1/L2 wide-lane (Equation (3)) this yields two independent band-difference constraints,

$$\delta N_{\mathrm{WL}} = \delta N_1 - \delta N_2, \qquad \delta N_{\mathrm{EWL}} = \delta N_2 - \delta N_5, \tag{5}$$

narrowing the per-signal triple $(\delta N_1, \delta N_2, \delta N_5)$ to a single remaining degree of freedom [2].

1.2.3 GRAPHIC (single-frequency half-sum)

The GRAPHIC combination averages the code and phase on a *single* band (`GraphicSlipDetector::computeGraphic`):

$$\mathrm{GR}^i_s \;=\; \tfrac{1}{2}\big(P^i_s + \Phi^i_s\big), \tag{6}$$

with Φ^i_s in metres. Because the slant ionosphere enters $+\mu_s I^i_{r,1}$ on the code and $-\mu_s I^i_{r,1}$ on the phase of the *same* signal (the uncombined code and phase observation equations), the half-sum cancels it to first order — GRAPHIC is ionosphere-free without a second frequency, at the cost of retaining half the code noise. The combination still contains the geometric range, clocks and troposphere plus the constant half-ambiguity $\frac{1}{2}\lambda_s N^i_s$, so the between-epoch first difference

$$\mathrm{dGR}^i_s(k) \;=\; \mathrm{GR}^i_s(k) - \mathrm{GR}^i_s(k-1) \tag{7}$$

removes the constant ambiguity (and cancels range and clocks down to receiver-motion / code-noise level). A δN-cycle slip on signal s therefore produces a jump of $\frac{1}{2}\lambda_s\,\delta N$, and the integer estimate is $\delta N = \mathrm{round}\big(2\,\mathrm{dGR}/\lambda_s\big)$. With $\sigma_P \sim 0.3\,\mathrm{m}$ the post-difference noise is $\sim 0.3\,\mathrm{m}$, so sensitivity is noise-limited to a few L1 cycles; in practice a hard floor `kMinThresholdM` $= 0.5\,\mathrm{m}$ governs on quiet arcs (≈ 5 L1 cycles), keeping GRAPHIC a coarse backstop rather than a sub-cycle detector [1, 3].

1.2.4 Geometry-free phase (staged)

The geometry-free (GF) observable is the between-band phase difference

$$\mathrm{GF}^i \;=\; \Phi^i_1 - \Phi^i_2 \;\approx\; (\mu_2 - \mu_1)\, I^i_{r,1} \;+\; (\lambda_1 N^i_1 - \lambda_2 N^i_2), \tag{8}$$

which removes geometry, clocks and troposphere exactly but retains the *difference* of the two ionospheric delays. Since $I^i_{r,1}$ varies smoothly, a second-order time difference

$$\mathrm{TD2}_{\mathrm{GF}}(k) \;=\; \mathrm{GF}^i(k) - 2\,\mathrm{GF}^i(k-1) + \mathrm{GF}^i(k-2) \tag{9}$$

cancels the ionospheric trend up to quadratic order over the sampling interval, leaving noise plus slip (`GfSlipDetector::computeGfMeters` for Equation (8); the TD2 update is the level-jump test statistic). The detector is intrinsically high-rate ($\geq 1\,\mathrm{Hz}$): at coarse rate the residual ionospheric curvature defeats (9). It is present in the tree but excluded from the build; the same algebra backs the active `src/core/slip/IonoVariabilityMonitor.cpp`.

1.2.5 Ionosphere-free phase (staged)

The dual-frequency ionosphere-free (IF) phase combination uses the coefficients

$$c_1 = \frac{f_1^2}{f_1^2 - f_2^2}, \qquad c_2 = \frac{f_2^2}{f_1^2 - f_2^2}, \qquad c_1 - c_2 = 1, \tag{10}$$

giving $\Phi^i_{\mathrm{IF}} = c_1\Phi^i_1 - c_2\Phi^i_2$ and the effective wavelength $\lambda_{\mathrm{IF}} = \left|c_1\lambda_1 - c_2\lambda_2\right|$ (`IfPhaseSlipDetector::computeIfWavelengthMeters`); for GPS L1/L2 $\lambda_{\mathrm{IF}} \approx 0.107\,\mathrm{m}$. Rather than screen the raw combination, the staged detector forms the IF observed-minus-computed (OMC), differences it between satellites against a high-elevation pivot p_g to remove the receiver clock, the inter-system offset $c\,\delta t_r^{(g)}$ and the satellite phase clock, then between epochs to remove the slowly-varying wet troposphere and the constant ambiguity:

$$\nabla\Delta\Phi^{i,p_g}_{\mathrm{IF}}(k) = \left[\Phi^i_{\mathrm{IF}}(k) - \Phi^{p_g}_{\mathrm{IF}}(k)\right] - \left[\Phi^i_{\mathrm{IF}}(k-1) - \Phi^{p_g}_{\mathrm{IF}}(k-1)\right], \tag{11}$$

which is near zero for a clean arc and jumps by $\sim \delta N\,\lambda_{\mathrm{IF}}$ on a slip. The between-satellite step does not cancel the differential geometric error $(\boldsymbol{e}^i - \boldsymbol{e}^{p_g})^{\mathrm{T}}\delta\boldsymbol{r}$, so flagging must be gated by the filter's 3D position sigma; this restricts the staged detector to a static site with a precise a priori position, which is why `estimation.enable_if_slip_detector` is currently inert.

1.3 TurboEdit dual-frequency resolution

The TurboEdit resolver is a *pure* MW+GF joint integer solver (`src/core/slip/TurboEditResolver.cpp` and its header `include/slip/TurboEditResolver.hpp`). It owns the per-arc geometry-free jump history and the integer arithmetic, and returns per-signal integer repairs plus a reset-fallback set without mutating filter state; it is wired into both the post-processing `CycleSlipManager` and the real-time engine, and enabled by `estimation.cycle_slip.turbo_edit`.

1.3.1 The 2×2 system

For a dual-frequency phase pair (s_1, s_2) on satellite i, ordered higher-frequency first ($f_{s_1} > f_{s_2}$, hence $\lambda_{s_1} < \lambda_{s_2}$), the resolver pairs two scalar slip observables. The first is the MW wide-lane integer of Equation (3), taken *as input* from the MW detector and never recomputed (the empirically tuned MW path of Section 1.2.1 is the single source of δN_{WL}):

$$\delta N_{\mathrm{WL}} \;=\; \delta N_{s_1} - \delta N_{s_2} \qquad \text{(one equation, two unknowns).} \tag{12}$$

The second is the GF phase jump in metres. Writing the GF level as $\mathrm{GF}(t) = \lambda_{s_1} L_{s_1}(t) - \lambda_{s_2} L_{s_2}(t)$ (carrier in cycles), a simultaneous slip steps its ambiguity content by

$$\delta\mathrm{GF} \;=\; \lambda_{s_1}\,\delta N_{s_1} - \lambda_{s_2}\,\delta N_{s_2}, \tag{13}$$

estimated in-arc by the second-order time difference $\mathrm{TD2}(t) = \mathrm{GF}(t) - 2\,\mathrm{GF}(t-1) + \mathrm{GF}(t-2)$ of Equation (9), which cancels the ionospheric delay up to quadratic order so that at the slip epoch it equals the integer jump plus phase noise only. The TD2 chain requires three *consecutive* epochs; non-consecutive history forces a chain re-seed (Section 1.3.4).

Stacking Equations (12) and (13) gives the linear map from the integer slip pair to the two observed jumps:

$$\boxed{\begin{pmatrix} \delta N_{\mathrm{WL}} \\ \delta \mathrm{GF} \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & -1 \\ \lambda_{s_1} & -\lambda_{s_2} \end{pmatrix}}_{\boldsymbol{A}} \begin{pmatrix} \delta N_{s_1} \\ \delta N_{s_2} \end{pmatrix}.} \tag{14}$$

The determinant $\det \boldsymbol{A} = \lambda_{s_1} - \lambda_{s_2}$ is non-zero for any genuine cross-band pair — it is the cm-scale quantity `denom` of Equation (16), $\approx -0.0539\,\mathrm{m}$ for GPS L1/L2, *not* the $\sim 0.86\,\mathrm{m}$ wide-lane wavelength — so the system is always invertible:

$$\boldsymbol{A}^{-1} = \frac{1}{\lambda_{s_1} - \lambda_{s_2}} \begin{pmatrix} -\lambda_{s_2} & 1 \\ -\lambda_{s_1} & 1 \end{pmatrix}. \tag{15}$$

The implementation does not invert the full matrix; it solves the second row for δN_{s_2} and back-substitutes the first (`TurboEditResolver.cpp`):

$$\delta N_{s_2} = \frac{\delta \mathrm{GF} - \lambda_{s_1}\, \delta N_{\mathrm{WL}}}{\lambda_{s_1} - \lambda_{s_2}}, \qquad \delta N_{s_1} = \delta N_{s_2} + \delta N_{\mathrm{WL}}, \tag{16}$$

with δGF instantiated either by the in-arc TD2 or, across a data gap, by the bridged jump of Section 1.3.4. The denominator $\lambda_{s_1} - \lambda_{s_2}$ (the variable `denom` in the source) equals $\approx -0.0539\,\mathrm{m}$ for GPS L1/L2, so the amplification factor on the wide-lane integer is $\lambda_{s_1}/(\lambda_{s_1} - \lambda_{s_2}) \approx -3.53$: the GF equation must localise δN_{s_2} to well within ± 0.5 cycle despite that lever, which is what the split-capability gate below enforces.

1.3.2 Integer rounding and validation

The float δN_{s_2} from Equation (16) is rounded and its residual recorded:

$$d_2 = \mathrm{round}\big(\delta N_{s_2}\big), \quad f = \big|\delta N_{s_2} - d_2\big|, \quad d_1 = d_2 + \delta N_{\mathrm{WL}}. \tag{17}$$

A candidate is *accepted* as a repair only if the rounding is clean ($f \le 0.30$, the constant `kMaxFractional`; a $1 - 0.30 = 0.70$ cycle margin to the nearest wrong integer), the magnitudes are sane ($|d_1|, |d_2| \le$ `kMaxSlipCycles`), and the pair is non-trivial ($d_1 \neq 0 \vee d_2 \neq 0$). A non-trivial accepted candidate yields `action="repair"`; $(d_1, d_2) = (0, 0)$ is resolved as no-slip and the arc keeps rolling; otherwise the satellite is reset.

Before the joint solve is *trusted*, the GF evidence must be sharp enough to discriminate the integer lattice. The float δN_{s_2} carries $\sigma_{\mathrm{TD2}}/|\lambda_{s_1} - \lambda_{s_2}|$ cycles of noise, so the solve is permitted only when

$$\sigma_{\mathrm{TD2}} \le \texttt{turbo_edit_split_max_sigma} \cdot \big|\lambda_{s_1} - \lambda_{s_2}\big| \qquad (\texttt{splitCapable}), \tag{18}$$

with σ_{TD2} the arc's own MAD-derived σ when mature, else the cross-satellite tier floor, else NaN (NaN fails by design — "no estimate" is not evidence of sharpness). The default `estimation.cycle_slip.turbo_edit_split_max_sigma` is 0.2 cycles. This gate is the single mechanism making the resolver rate-adaptive: millimetric 1 s noise passes trivially, while coarse-rate (30 s) ionospheric residual passes only while the arc is quiet. On an MW-flagged pair failing the gate the resolver falls back to a satellite reset; a GF-only exceedance on a split-incapable arc is *ignored*, not reset. When $\delta N_{\mathrm{WL}} = 0$ (an equal slip $\delta N_{s_1} = \delta N_{s_2}$, to which MW is blind) a GF-only flag fires on $|\mathrm{TD2}|$ exceeding an adaptive threshold $\tau = \max(k_\sigma \hat{\sigma}_{\mathrm{TD2}}, 0.04\,\mathrm{m})$, with k_σ the `turbo_edit_sigma_multiplier`; the floor sits just below $|\lambda_{s_1} - \lambda_{s_2}| \approx 0.054\,\mathrm{m}$ so a ± 1-cycle equal slip remains detectable.

1.3.3 MW-only / code-event null line

The asymmetric case is MW flagged but the GF jump *affirmatively zero*. For a fixed $\delta N_{\mathrm{WL}} = k$, every integer pair $(\delta N_{s_1}, \delta N_{s_1} - k)$ predicts, via Equation (13), a GF jump

$$\delta\mathrm{GF} \;=\; \lambda_{s_2} k - (\lambda_{s_2} - \lambda_{s_1})\, \delta N_{s_1}, \tag{19}$$

whose zero crossing sits at $\delta N_{s_1} = \lambda_{s_2} k / (\lambda_{s_2} - \lambda_{s_1}) \approx 4.53\, k$ for GPS L1/L2 ($\approx 3.95\, k$ for L1/L5). Every small pair $|\delta N_{s_1}| \le 3$ then predicts $|\delta\mathrm{GF}| \ge 8\,\mathrm{cm}$ — orders above phase noise — so a flat GF leaves only *null-line* pairs of $(4, 3)$ / $(9, 7)$-class magnitude: implausible multi-cycle simultaneous slips on both signals with no loss of lock. The Occam-simpler explanation is a narrow-lane *code* excursion (multipath, NLOS, DLL re-lock), since MW $= \Phi_{\mathrm{WL}} - P_{\mathrm{NL}}$ and one wide-lane cycle $\approx 0.86\,\mathrm{m}$ of narrow-lane code.

Accordingly the pair is classified a code-domain artifact — and the caller *skips* the phase reset — when the GF jump is affirmatively zero within a discrimination-bounded gate,

$$|\mathrm{TD2}| \le \underbrace{\texttt{kCodeEventZeroSigma} \cdot \hat{\sigma}_{\mathrm{MAD}}}_{\text{zero gate}} \quad\wedge\quad \text{zero gate} \le \underbrace{\texttt{kVetoZeroGateMaxLambdaFrac} \cdot |\lambda_{s_1} - \lambda_{s_2}|}_{\text{discrimination cap}}, \tag{20}$$

with `kCodeEventZeroSigma` $= 3.5$ ($\approx 99.95\%$ two-sided) and `kVetoZeroGateMaxLambdaFrac` $= 0.5$. The cap is the rate-adaptive safeguard: a zero gate wider than half the smallest GF-visible quantum cannot distinguish stillness from a small slip, so the veto stands down and the MW mandate governs. The classification is further trust-gated — veto enabled, MW flagged, GF not exceeding, no post-slip cooldown, ionospherically stable constellation, no LLI on either signal, and MAD maturity ≥ 10 samples (stricter than detection's 5, because affirming stillness is a state-*preserving* decision). The implied null-line pair is computed for the trace only, never applied; `did_slip` stays false so the GF chain keeps rolling. A genuine null-line slip is not lost: it jumps that signal's Kalman phase innovation by $\ge |4\lambda_{s_1}| \approx 0.76\,\mathrm{m}$ this very epoch, far beyond any residual gate, so the residual-driven reset corrects it one epoch late. The behaviour is toggled by `estimation.cycle_slip.turbo_edit_mw_only_veto` (default on).

1.3.4 Gap-spanning GF bridge

The TD2 chain needs three consecutive epochs; any per-signal dropout or receiver gap re-seeds the chain, making the post-gap TD2 *blind* to a re-lock slip whose GF level jump lives *between* the two chains. When a chain break occurs within `estimation.cycle_slip.turbo_edit_max_gap_s`, the resolver stashes the pre-gap GF level $\mathrm{GF}_{\mathrm{ref}}$, its timestamp, the arc's EWMA GF rate $\dot{\mathrm{GF}}$ and the pre-gap spacing dt_0, and on return forms the drift-corrected across-gap jump

$$\boxed{\delta\mathrm{GF}_{\mathrm{gap}} \;=\; \mathrm{GF}_{\mathrm{now}} - \mathrm{GF}_{\mathrm{ref}} - \dot{\mathrm{GF}}\,\Delta t, \qquad \Delta t = t_{\mathrm{now}} - t_{\mathrm{ref}},} \tag{21}$$

which then plays the role of $\delta\mathrm{GF}$ in Equation (16). Here $\dot{\mathrm{GF}}$ is an EWMA ($\alpha = 0.1$) of the per-epoch GF rate, chosen so its noise does not amplify when projected over the gap. Because the un-cancelled ionospheric residual is random-walk over the gap, the detection threshold and the code-event zero gate are widened by the factor $w = \sqrt{\Delta t / dt_0}$ (for $\Delta t > dt_0$), and the split-capability σ becomes $\sigma_{\mathrm{bridge}} = \hat{\sigma}_{\mathrm{MAD}}\, w$; the zero-gate *discrimination cap* stays un-widened, since the integer lattice of Equation (14) does not stretch with the gap. The bridge cap default is 15 s in code, raised to 60 s in the bundled real-time and replay configs.

An MW-flagged pair across the gap is attributed three ways. (i) A *code event* (`action="code_event"`) when $\delta\mathrm{GF}_{\mathrm{gap}}$ is affirmatively zero within the widened gate — the same null-line logic as

Equation (20), applied across the gap — preserving phase through a re-acquisition DLL transient. (ii) A *repair* or *moot* when the joint solve runs with $\delta\mathrm{GF} = \delta\mathrm{GF}_{\mathrm{gap}}$; per signal it is dispatched by observation-domain continuity, a signal observed at every epoch through the gap kept its ambiguity (its nonzero δN is a genuine *repair*), whereas a signal that missed an epoch was pruned-and-reseeded on return (its nonzero δN is *moot* — already absorbed, mandate suppressed). (iii) A *reset* when the across-gap σ is too coarse for the split, or the candidate set fits zero or two integers with no deferral.

Because the GF equation alone is degenerate along the null-line lattice of Equation (19) — $+0.735\,\mathrm{m}$ fits both $(0, -3)$ and $(9, 4)$ at millimetre level; only the integer k separates them — the bridged solve demands a *unique* surviving candidate after three guards: the fractional gate of Equation (17); a GF-separation guard requiring the candidate's own predicted jump to clear the zero gate, $|\lambda_{s_1} d_1 - \lambda_{s_2} d_2| > \texttt{zero_gate}$, lest a marginal noise exceedance mint a multi-cycle repair from noise; and a sibling-pair zero affirmation, whereby a satellite's *other* GF arc that stayed chain-continuous and unflagged through the bridged interval affirms $\delta N = 0$ for both its signals and prunes any candidate demanding a nonzero integer on an affirmed signal. The wide-lane integer feeding the bridged solve is taken in trust order: the published MW integer k; failing that, at the flag epoch, $\mathrm{round}(f) \pm 3$ around the (NL-code-biased) instantaneous re-acquisition float; and on deferral epochs the settled across-gap MW Hatch mean. When the float wide-lane evidence is unsettled or the lattice is ambiguous (two or more acceptable integers), the solve is *deferred* rather than guessed (`estimation.cycle_slip.turbo_edit_wl_settle_max_s`), terminating in a later repair, reset or timeout-reset.

1.3.5 Atomic reconciliation

Per-satellite candidates are reconciled atomically: a cross-pair integer disagreement on a shared signal demotes the whole satellite to reset, and a satellite ending in the reset set has its code-event, moot, clean and pending classifications *withdrawn* — contradictory evidence always resolves to the conservative outcome. Arc bookkeeping keeps the TD2 validity precondition honest: a non-consecutive update partially resets the chain but preserves the slowly-varying MAD calibration window, an absolute gap beyond the configured maximum wipes the arc, and an accepted slip re-seeds with a short cooldown to suppress the TD2 echo; only clean (non-exceeding, non-bridged) TD2 samples feed the MAD window.

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